# ABHRANKAN CHAKRABARTI

Mathematics • Algorithms • Programming • Computational Science
abhrankan@gmail.com | +91-6289194286 | Kolkata, West Bengal, India

github.com/Abhrankan-Chakrabarti | LinkedIn | @AbhrankanC | abhrankan.dpdns.org

## EDUCATION

**Bachelor of Technology – Computer Science & Engineering** — **CGPA: 7.69** (Till 6th Semester)

*Techno Bengal Institute of Technology, Kolkata (MAKAUT)* — *2023 – 2027 (Expected)*

| Sem 1 | Sem 2 | Sem 3 | Sem 4 | Sem 5 | Sem 6 |
|---|---|---|---|---|---|
| 7.63 | 8.00 | 7.73 | 7.33 | 7.75 | 7.68 |

## TECHNICAL SKILLS

**Languages:** C, C++, Java, Python, TypeScript, JavaScript

**Web & Backend:** HTML, CSS, React (Vite), Node.js, Express.js, REST APIs

**Databases:** MongoDB (aggregation pipelines, ACID transactions)

**AI / ML Libraries:** NumPy, Pandas, scikit-learn

**AI / ML APIs & Inference:** Gemini Pro Vision, DALL-E 3, Stable Diffusion XL, llama.cpp

**Core Areas:** Algorithms, Number Theory, Numerical Methods, Cryptography, Computational Mathematics

**Tools & Practices:** Git, GitHub, Linux, VS Code, Jest, ESLint, MVC Pattern, Dependency Injection

## EXPERIENCE

**Winternship 2025 – Full Stack Development Trainee** — **2025**

*Ajrasakha Hackathon / Online Program* — *Remote*

- Completed structured case studies across the full MERN stack: TypeScript, React (Vite), MongoDB, and Express.js.
- Built and deployed two live React apps (Task Manager, Component App) hosted on Render.
- Implemented ACID-compliant MongoDB transactions, aggregation pipelines, and CRUD workflows.
- Developed modular REST APIs in Express with middleware chains, MVC pattern, and dependency injection.
- Applied design patterns (Observer, Strategy), generics, and advanced TypeScript types across 19+ lessons.

## PROJECTS

**Almost Isosceles Pythagorean Triples Optimizer**

*github.com/Abhrankan-Chakrabarti/almost-isosceles-pythagorean-triples*

Efficient generation and optimization of almost-isosceles Pythagorean triples using number-theoretic insights. Published an accompanying article on the mathematical exploration.

**Catalan $\pi$ Algorithm**

*github.com/Abhrankan-Chakrabarti/pi-calculation-with-catalan-numbers*

Mathematical exploration of $\pi$ computation using Catalan-number-based formulations and high-precision numerical optimization techniques.

**Cryptify – Encryption Tool**

*github.com/Abhrankan-Chakrabarti/Cryptify*

A cryptography-focused project implementing advanced encryption algorithms, exploring secure computation design and cybersecurity principles.

**VisionaryAI (GeminiFusion)**

*github.com/Abhrankan-Chakrabarti/GeminiFusion*

Web application leveraging Gemini Flash, DALL-E 3, and Stable Diffusion XL for chatbot interaction, image captioning, and AI image generation. (6 stars on GitHub)

**LlamaInteract**

*github.com/Abhrankan-Chakrabarti/LlamaInteract*

Local large language model interaction system built using llama.cpp for efficient on-device inference, enabling private and resource-conscious LLM usage.

**Interactive Graphing Calculator**

*github.com/Abhrankan-Chakrabarti/GraphingCalculator*

Dynamic graphing and visualization tool for mathematical functions and computational analysis.

**Complex Calculator (C++ & Python)**

*github.com/Abhrankan-Chakrabarti/ComplexCalculator*

Menu-driven modular calculator supporting complex mathematical operations, implemented in both C++ and Python.

**Prime Grid Visualizer**

*github.com/Abhrankan-Chakrabarti/prime-grid-visualizer*

Fast, interactive browser-based tool to explore and visualize prime numbers. (5 stars on GitHub)

## ARTICLES & WRITING

- Calculating π with Catalan Numbers
- Almost Isosceles Pythagorean Triples
- The Golden Ratio
- History of Cryptography

*Published on personal blog: abhrankan.dpdns.org*

## CERTIFICATIONS

**NPTEL – The Joy of Computing Using Python** | Elite + Gold | Score: 91% | Top 1% (Topper)

IIT Madras / Swayam NPTEL certification (Jan–Apr 2024, 12-week course). Ranked in the Top 1% among 12,432 certified candidates. Online Assignments: 24.06/25 | Proctored Exam: 66.97/75 | Roll No: NPTEL24CS57S559700017

## AREAS OF INTEREST

Artificial Intelligence • Mathematical Computing • Algorithms & Optimization • Number Theory • Cryptography • Scientific Computing • Numerical Methods

## HOBBIES & ACTIVITIES

Coding • Solving mathematical problems • Learning new technologies • Recitation • Singing